



CSE461

Introduction to Robotics Lab

Real-World Implication

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Introduction

The proposed **Smart Hospital Assistant Robot for Autonomous Medicine Delivery** is a robot which autonomously delivers medical supplies to patients inside the hospital. It delivers medications from a medicine storage to designated patient beds. The prototype is made of ESP32 development board (ESP32-CAM), IR line-following sensor, HC-SR04 ultrasonic sensor, DC gear motor, L298N motor driver, SG90 servo motor, buzzer, LED, LCD display, and a lockable medicine compartment. QR Code processing is done by a PC. The project thus combines embedded systems, computer vision, autonomous navigation and basic access control in a healthcare environment.

Purpose

The main goal of this project is to design and develop a hospital assistant robot that can safely transport medicine from a medical storage station to the patient's beds. Medicine delivery is a repetitive task where healthcare workers have to repeatedly move from medicine storage areas to patient locations. The robot can be used to automate the transportation part of this process, which will alleviate the need for unnecessary tasks to be carried out by healthcare workers and free up time for them to dedicate to patient care. Another goal of the project is to minimize human mistakes in the routine delivery of medicines. Medicine storage station and patient bed using QR code verification. This will be an extra layer of security before the medicine is sent and dropped. One of the major roles is medicine security. The servo-controlled compartment is closed when the medicine is inserted into the robot by the hospital staff. Locks the compartment when transported and unlocks only after successful patient or destination verification. The project showcases the seamless application of autonomous navigation, obstacle avoidance, computer vision, QR code recognition, embedded control and automated access control in a medical setting.

Societal implications

The potential future hospital assistant robot has a number of positive implications for society, particularly in the medical field in which medicines are given repeatedly in the healthcare facilities. The first is that the robot can optimize hospital operations. Healthcare workers will often need to walk from medicine storage areas to patient beds to deliver routine medicines. This process of transportation can be automated, which will decrease unnecessary movements and enable employees to concentrate on more significant patient care duties. Secondly, the robot can reduce some of the delivery errors made by humans. QR code verification is an additional verification step for the medicine requested and the intended patient bed. The robot is not a replacement for clinical verification, but it can help to reduce some errors in the physical delivery process. The system can also help to reduce the normal workload of health care workers. A robot that can repeat the same transportation task repeatedly could be helpful in hospitals where there are numerous patients and frequent deliveries of medicines. But, automation can also raise issues about jobs and evolving roles of health care professionals. The proposed robot should therefore be considered as an assistant, rather than a substitute for hospital staff. Staff would remain responsible for preparing medicines, checking medicines, loading medicines, exceptions and the management of the delivery of medicines. The system could be adapted to be used in hospitals, clinics, nursing homes, and other health care facilities. However, each facility would have to have the right route planning, safety measures and integration into existing workflow.

Health implications

The robot in the project reduces human interference, ensuring lower risk of infection and disease spreading, which is vital in isolation wards. QR code system scans the upcoming medicine through the robot, ensuring no wrong medicine or mix-ups are being supplied. It also operates continuously, becoming more efficient than humans, preventing any delays to the patient's treatment needs. However, it does not verify expiry dates or medicine amounts, so this depends on the carefulness of the doctor or pharmacists.

Safety implications

Using HC-SR04, the robot can detect obstacles in its route when going from its current position to its destination. This prevents collision of the robot against real-life obstacles like moving humans, medical equipment, walls, etc. The robot keeps the medical compartment carrying medicines locked, which ensures the safety of the medicine. It cannot be stolen, switched and tampered with during its route. Only when matched with the correct QR code, the compartment will unlock itself. However, if its battery runs out then it may cause blockage between the hospital rooms, requiring human intervention. Ultrasonic sensing cannot detect high-moving obstacles and low-lying objects, which can be a problem in real-life scenarios.

Legal implications

The robot needs to safeguard patient and prescription data accessed via QR codes and the PC system. Only authorized hospital personnel should be allowed to access. Clearly defined accountability for medicine preparation, failures in the operation and delivery of the robot. The system should also follow applicable healthcare, safety, privacy and cybersecurity regulations. The robot is used to deliver medicines, therefore the correct procedures need to be followed to ensure the right medicine is delivered to the right patient. If there is any delivery mistake it should be noted and dealt with by authorised personnel. The prototype is not a clinically approved medical device. A real-life version would need to be tested, secure, supervised by humans and adhere to the relevant laws and hospital policies.

Cultural implications

While the introduction of robots in hospitals could be seen as a contemporary and effective solution, some patients and healthcare workers may be hesitant or uncomfortable with the technology at first. The robot should support, not replace, healthcare workers. Staff should be on hand to support patients who might have problems using or interacting with the system.

The LCD, LEDs and buzzer can communicate clearly and help establish trust in the system to the user, letting them know the status of the robot. The robot should also be simple to operate and flexible to various hospital settings, languages and procedures. Training and staff engagement can help to ensure effective and successful adoption of the technology. The robot needs to be user-friendly and flexible to various hospital settings, languages and processes. Proper training and staff involvement can help ensure that the technology is accepted and used effectively.

Conclusion

The Smart Hospital Assistant Robot is an example of the use of robotics, automation and computer vision in the routine delivery of medicine in healthcare facilities. The robot can autonomously navigate and detect obstacles, verify the QR code, and transport the medicine to the correct designated patient bed and return to the medicine station safely, thanks to the combination of autonomous navigation, obstacle detection, QR code verification and a secure medicine compartment. The system can help to decrease the repetitive workload of healthcare workers, increase the efficiency of delivering healthcare services, decrease some human errors, and increase the security of medicines. But, there is still a need for human supervision in medicine preparation, verification and handling unexpected situations. The robot should be used as a health care worker assistant and not a replacement. The project could serve as a good starting point for future autonomous healthcare assistance robots if the required safety tests are done and the hospital approves the robot for use.